

Near-Realtime Knowledge Update on the Legal Search Platform

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Abstract—Updating legal information is crucial to ensure the accuracy and timeliness of legal regulations. This study introduces a platform capable of updating legal information based on the latest legal documents to swiftly assist users in accessing new regulations during legal inquiries. The platform automatically identifies newly enacted content, determines update requirements and changes from these documents, maps specific outdated content needing replacement, and updates legal knowledge through corresponding knowledge graph updates. Experiments have been conducted, yielding positive results to demonstrate the feasibility of this approach.

Index Terms—Legal Information, Knowledge Graph, Ontology, Knowledge-based systems, Knowledge Update.

I. INTRODUCTION

Today, AI technology has been widely applied in the legal field, bringing significant benefits [5]–[7]. In addition, intelligent legal query systems can support legal decision-making quickly and accurately. These systems help reduce the workload for humans, while also improving efficiency and reliability in handling legal issues [4]. However, queries and decisions must always be based on the most recent legal documents. This requires legal information to be continuously updated to avoid obsolescence and inaccurate results.

The platform of this study uses a knowledge graph as the knowledge base for the system [12], [13]. Legal documents are organized and represented as a knowledge graph. The questions from end-users are converted into graphs that represent the semantics of the questions. The platform then searches for subgraphs in the knowledge base that match the knowledge graph of the questions to provide answers to the users [13], as illustrated in Figure 1.

In reality, legal documents are constantly changing, updating, and improving to keep up with societal developments and the needs of the people. The process of updating legal information is diverse, potentially involving the issuance of a new document that replaces the entire content of the old one, or simply adjusting, amending, or supplementing specific parts. This creates many challenges and difficulties for the process of updating legal information, which can be time-consuming and labor-intensive for humans.

Updating legal documents often encounters the following situations. Firstly, a new document is issued to completely replace the content of an old document, such as Decree 100/2019/ND-CP¹ completely replacing Decree 46/2016/ND-CP². In these cases, the platform does not need to identify detailed changes for the update but will delete the entire content of the old document and represent the new document to replace it in the knowledge base.

This research introduces a platform capable of supporting the updating of legal documents when new documents are issued. This platform not only helps accurately identify the parts of the content that need to be replaced but also updates the information quickly and accurately. In the legal system in Vietnam, new legal documents are often issued to adjust, amend, or supplement certain points in the old document without replacing the entire content. This requires the support platform to have the ability to accurately process and recognize these changes to ensure the legal information is updated efficiently and promptly.

II. RELATED WORK

Nowadays, to meet the continuous demand for information updates across various fields, research on knowledge updating has been conducted [3], yielding several notable results as follows:

[15] proposed a method for updating embeddings in dynamic Knowledge Graphs (KGs). Those graphs undergo frequent changes with the addition or removal of nodes and edges. Instead of recalculating embeddings from scratch after each minor change, the study focuses on updating only the embeddings related to new changes. This approach saves time and resources while maintaining the quality of Knowledge Graphs as they continue to evolve. However, effective implementation of this method requires substantial historical data on KG changes.

Similarly, the authors in [2] built an AIR framework to adaptively update embeddings for dynamic KGs. This method measures the importance score of each triple in the KG

¹Decree No. 100/2019/ND-CP on Administrative penalties for road traffic and rail transport offences, 2019

²Decree No. 46/2016/ND-CP on administrative penalties for road traffic offences and rail transport offences, 2016

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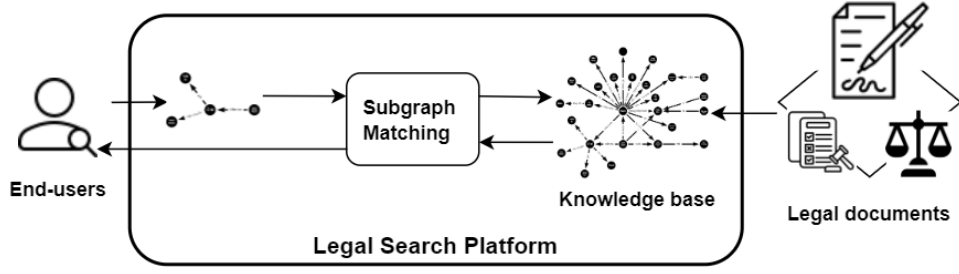


Fig. 1. Legal Search Platform Overview.

and, based on this score, AIR selects a subset of triples most affected by new information for updating. Besides, AIR updates the embeddings without needing to retrain the entire model from scratch, thus saving time and resources. However, the effectiveness of AIR depends on the accuracy of the importance scores assigned to each triple and may face challenges when handling large KGs.

The study in [19] discussed a method that uses exceptions to modify existing rules instead of directly changing those rules. This approach helps maintain the integrity of the foundational knowledge and can be more efficient and manageable compared to traditional knowledge update methods. By using RE models, this method provides a more suitable semantic foundation for rule updates. However, it is a highly complex method and poses challenges in accurately identifying and representing exceptions.

The study in [9] proposed a new method, called StableKE, for efficiently updating knowledge in large language models (LLMs). This method utilizes Semantic Paraphrase Enhancement and Contextual Description Enrichment strategies to augment knowledge. However, the effectiveness of this method heavily relies on the quality of the data used for training and evaluation. Moreover, StableKE does not provide detailed information on the decision-making process.

[10] presented the UnKE method for updating unstructured knowledge in large language models. This method uses key-value pairs at the layer level to represent and edit unstructured knowledge effectively and flexibly. However, this method depends on the architecture of the LLM and requires adjustments when applied to new LLM models. Additionally, it lacks transparency and has limitations in terms of explainability.

Automating the process of updating legal information can help monitor, analyze, and update changes in legal documents, thereby supporting legal professionals and the public in accessing information easily and accurately. This not only saves time and effort but also improves the effectiveness of applying and complying with the law.

Existing methods for automatic and rapid knowledge updating often prove inadequate or require substantial implementation effort. This is particularly evident in the domain of legal knowledge, where research on knowledge updates is limited, and adapting methods from other fields is challenging. This study focuses on incremental updates triggered by the issuance

of new documents that amend or supplement existing content. The proposed platform aims to identify specific changes within these new documents and seamlessly integrate them into the corresponding knowledge graph, thereby ensuring accurate and timely information representation.

III. KNOWLEDGE UPDATE

A. Text Analysis Techniques

To analyze the content and the required changes in the legal documents, this research develops a unique method to analyze updated documents, aiming to identify the requirements and determine the positions that need to be changed.

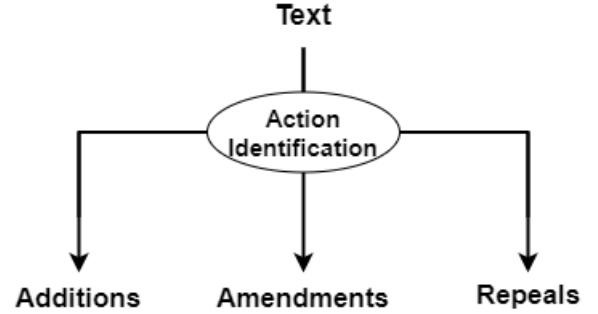


Fig. 2. Simulation of the Action Identification Process.

1) *Identifying the Set of Words Describing Document Modification Actions:* Identifying the main action of a document item, as illustrated in Figure 2, is a crucial step in understanding the implications of changes in legal texts. The specific steps include:

a. Collecting Key Phrases

- **Data Source:** Collect key phrases that are the main verbs in sentences from the Amending and Supplementing Law on the Promulgation of Legal Normative Documents. Data sources are collected from legal databases and government websites.

- **Extraction:** Extract key phrases related to actions from document items.

b. Grouping Key Phrases

- **Action Classification:** The collected key phrases will be grouped into different types of actions such as amendments,

additions, replacements, and repeals. This classification helps the system easily identify and process changes in legal texts.

Due to the precise language used in legal texts, the collection and classification of key phrases will achieve high accuracy. This reduces errors in identifying and processing changes. It helps the system quickly and accurately identify the actions required for legal texts. Grouping key phrases by action also improves the efficiency of the document updating process.

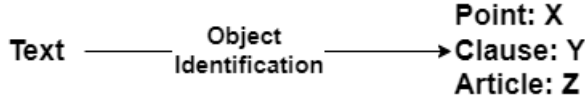


Fig. 3. Simulation of the Object Identification Process.

2) *Identifying Accompanying Objects*: Identifying the accompanying objects of the actions determined above or in other words, the specific positions in the old document that the document item refers to, as illustrated in Figure 3, is a crucial step to ensure that all changes are made accurately and consistently. The steps include:

- **Context Analysis**: Analyze the context of key phrases and surrounding content to ensure the identified position is accurate and relevant.

- **Determining Scope**: Identify the scope of the paragraph, clause, or point that will be affected by the change. This includes determining the relevant sections in the old document and ensuring that all changes are applied within the correct scope.

Example: a) Amend the opening paragraph of Clause 3, Article 44 as follows: "3. A fine ranging from VND 2,000,000 to VND 3,000,000 shall be imposed on the train driver, train captain, vice train captain responsible for safety for committing one of the following violations:"; The specific object to be amended will be "Clause 3, Article 44".

3) *Determining the Content of Changes*: Accurately determining the scope of changes required:

- **Context Analysis**: Analyzing the context helps the system better understand the implications of the changes and ensures that all changes are executed correctly.

Example: "23. Amend the opening paragraph and Point a of Clause 3 of Article 44 as follows: a) Amend the opening paragraph of Clause 3, Article 44 as follows: "3. A fine ranging from VND 2,000,000 to VND 3,000,000 shall be imposed on the train driver, train captain, vice train captain responsible for safety for committing one of the following violations:";" The change here only applies to the opening paragraph of the Clause, not the entire content of the Clause.

B. Analyzing Amendments

Analyzing the changes based on the newly issued legal documents is the crucial first step. To achieve this, we need to identify the different types of changes that may occur in Vietnamese legal texts. To accomplish this:

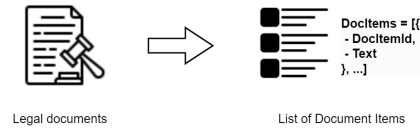


Fig. 4. Simulation of the Document Segmentation Process.

1) *Break down the new text into document items*:: Legal documents will be divided into document items, as shown in Figure 4. Document items are components within a document that represent specific content (such as a point or clause without containing subpoints) within that document.

2) *Identifying the Primary Actions of Document Items*: The sophisticated text analysis technique described in Section 3.A.1 is utilized to accurately identify the primary actions of the specified document items. These actions, which may include amendments, updates, and other modifications, are crucial for understanding the changes within the legal documents. Following identification, these actions are systematically categorized into types of changes, collected, and classified as follows:

Type 1: Content Modifications (Amendments)

This type of amendment involves making changes to the existing content of the legal document. It includes the actions described in Table I as follows:

TABLE I
SUMMARY OF CONTENT MODIFICATIONS

No.	Amendment Category	Description
1	Amendment of Articles	Modifying specific articles within the existing legal document to reflect new or updated information
2	Amendment of Clauses	Changing the content of specific clauses within a particular article to ensure clarity or accuracy
3	Amendment of Points	Modifying the content of specific points within a clause to update or correct the information
4	Replace Phrases	Substituting certain phrases within the document to reflect new terminology or correct errors
5	Remove Phrases	Deleting specific phrases from the document that are no longer relevant or accurate
6	Amendment of Article Titles	Changing the titles of specific articles to better represent the content or purpose of the article
7	Amendment of Clause Titles	Modifying the titles of specific clauses to improve clarity or accuracy
8	Amendment of Introductory Paragraphs	Changing the introductory paragraphs of specific clauses or articles to provide better context or clarity

Type 2: Content Additions (Additions)

This type of amendment involves adding new content to the legal document. It includes the actions described in Table II as follows:

Type 3: Content Repeals (Repeals)

This type of amendment involves removing content from the legal document. It includes the actions described in Table

TABLE II
SUMMARY OF CONTENT ADDITIONS

No.	Addition Category	Description
1	Addition of Points	Adding new points to the existing clauses to incorporate new information or regulations
2	Addition of Clauses	Incorporating new clauses into the existing articles to expand on or clarify existing information
3	Addition of Articles	Adding new articles to the legal document to introduce entirely new sections or topics

III as follows:

TABLE III
SUMMARY OF CONTENT REPEALS

No.	Repeal Category	Description
1	Repeal of Points	Removing specific points from the document that are outdated, irrelevant, or redundant
2	Repeal of Clauses	Eliminating specific clauses from the document that are no longer applicable or necessary
3	Repeal of Articles	Deleting entire articles from the legal document to streamline the content or remove obsolete sections

Example:

a) *Amendment of Point e, Clause 2, Article 28 as follows: "e) Not numbering seats, beds on passenger cars according to regulations;"*

Through the parsing technique, the research identifies that this document item falls under the category of "Amendment."

C. Mapping Changes

Mapping the changes from the new document to the old one is crucial for identifying specific items and content that need to be modified. This process involves:

- **Identifying Specific Items:** Accurately pinpointing the location of the content to be modified in the old document using the technique outlined in Section III-A2. This allows the system to retrieve the relevant content from the stored old document.
- **Scope of Changes:** Employing the method described in Section III-A3 to define the scope of the required changes based on the retrieved old content, ensuring that all modifications are fully and accurately updated.

Once these two steps are completed, the platform updates the new content in conjunction with the old content to create the most current version of the document.

Example:

a) *Amendment of the introductory paragraph of Clause 3, Article 44:*

"3. A fine ranging from 2,000,000 VND to 3,000,000 VND shall be imposed on train drivers, chief conductors, and deputy

chief conductors responsible for safety who commit one of the following violations:"

- **Specific Item:** The item to be amended is Clause 3, Article 44, with the old content retrieved from the system.
- **Scope of Changes:** The system identifies that the scope of the change is limited to the introductory paragraph of the clause, not the entire clause.

D. Updating the Knowledge Graph

The content of legal documents is stored in the form of knowledge graphs, where conceptual keyword clusters serve as nodes and the relationships between them are the edges of the graph. This knowledge graph forms the foundational knowledge base for the legal query system. The process of constructing a knowledge graph involves converting legal documents provided by users into knowledge graphs that represent the content of the legal documents. This process is illustrated in Figure 5 and has been presented in the study [12] The knowledge graph is composed of a set of triples that represent concepts and the relationships between them, facilitating faster and more accurate analysis and processing of information.

Updating legal information in the knowledge base involves updating the knowledge graph based on the changes that have been analyzed, performed according to various types. In other words, it involves updating the triples related to the changed content. This is carried out according to the following types of updates:

Type 1: Content Modifications (Amendments)

Content modifications involve making changes to the existing content in the knowledge base. This process is illustrated in Figure 6 and is carried out according to the following specific steps

KnowledgeBase = [ALL_TRIPLES_KNOWLEDGE_BASE]

• Step 1: Identify old triples

```
OldTriples = EMPTY_LIST
FOR Triple IN KnowledgeBase:
    IF ShouldReplace(Triple):
        ADD Triple TO OldTriples
RETURN OldTriples
```

• Step 2: Remove old triples

```
FOR Triple IN OldTriples:
    REMOVE Triple FROM KnowledgeBase
RETURN KnowledgeBase
```

• Step 3: Identify parent entities

```
ParentEntities = EMPTY_LIST
FOR Entity IN KnowledgeBase:
    IF IsParent(Entity, NewTriple):
        ADD Entity TO ParentEntities
RETURN ParentEntities
```

• Step 4: Implement changes

```
ADD NewTriple TO KnowledgeBase
RETURN KnowledgeBase
```

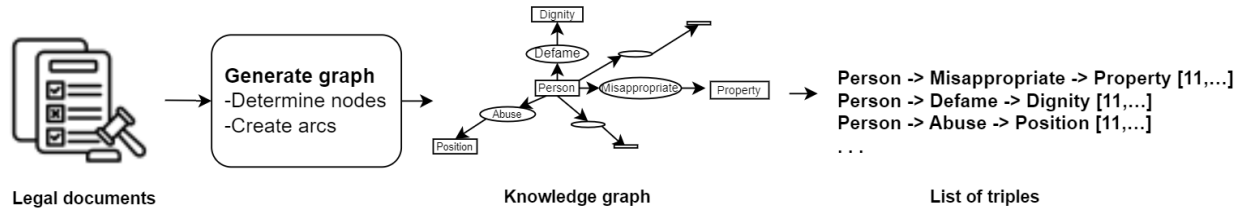


Fig. 5. Process of constructing a knowledge graph from legal documents.



Fig. 6. Simulation of the Graph Amendment Process.



Fig. 8. Simulation of the Graph Repeal Process.

Type 2: Content Additions (Additions)

Content additions involve integrating new information into the existing knowledge base, or in other words, adding new triples to the system. This process is illustrated in Figure 7 and is carried out through the following specific steps:



Fig. 7. Simulation of the Graph Addition Process.

• Step 1: Identify parent entities

```
ParentEntities = EMPTY_LIST
FOR Entity IN KnowledgeBase:
    IF IsParent(Entity, NewTriple):
        ADD Entity TO ParentEntities
RETURN ParentEntities
```

• Step 2: Implement changes

```
ADD NewTriple TO KnowledgeBase
RETURN KnowledgeBase
```

Type 3: Content Repeals (Repeals)

Content repeals involve removing information from the system's knowledge base, specifically by deleting the triples related to the content being removed. This process is illustrated in Figure 8 and is carried out through the following specific steps:

• Step 1: Identify related triples

```
RelatedTriples = EMPTY_LIST
FOR Triple IN KnowledgeBase:
    IF IsRelated(Triple, ContentIdentifier):
        ADD Triple TO RelatedTriples
RETURN RelatedTriples
```

• Step 2: Remove triples

```
FOR Triple IN RelatedTriples:
    REMOVE Triple FROM KnowledgeBase
RETURN KnowledgeBase
```

IV. EXPERIMENTAL RESULTS

To verify the effectiveness of the platform, the study conducted specific experiments with road traffic laws, an area frequently updated and revised. The chosen subject of study was Decree 123/2021/ND-CP³ (The updated content replacing Decree 100 on road traffic regulations.), issued to amend and supplement certain provisions of Decree 100/2019/ND-CP⁴ related to traffic violations and penalties. The experiment aimed to evaluate the platform's capability in identifying and analyzing change requests in new legal texts, as well as accurately updating these changes in the system's knowledge base, specifically its knowledge graph.

TABLE IV
DETAILS OF DECREE DECREE 100/2019/ND-CP AND DECREE 123/2021/ND-CP

	Decree 100/2019/ND-CP	Decree 123/2021/ND-CP
Chapter	5	1
Article	86	1
Clause	442	36
Point	1354	116

The platform analyzed the content of Decree 123, identifying change requests, additions, or deletions. Ultimately, the platform updated these changes in the system's knowledge

³Decree No. 123/2021/ND-CP on Amending and supplementing some articles of decrees provisions on penalties for administrative violations in the marine region, road, rail traffic, and domestic airline, 2021

⁴Decree No. 100/2019/ND-CP on Administrative penalties for road traffic and rail transport offences, 2019

graph, ensuring accurate and consistent integration of new information.

The experimental results demonstrated the platform’s success in identifying and analyzing change requests in Decree 123. Specifically, the platform identified 128 items requiring changes, including modifications, additions, and removals, totaling 340 points, sections, and articles. Detailed results are provided in Table V and Figure 9..

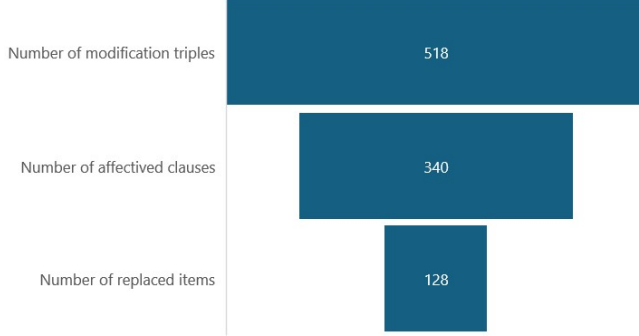


Fig. 9. The quantity of the platform’s updating in Decree 123.

TABLE V
SUMMARY OF DOCUMENT CHANGES

Change Type	Number of items requiring replacement	Number of affected items	Update Type
Amendment of Articles	9	156	Amendments
Amendment of Points	44	48	
Amendment of Clauses	22	62	
Replace Phrases	19	19	
Remove Phrases	4	4	
Amendment of Article Title	1	1	
Amendment of Clause Title	1	1	
Amendment of Introductory Paragraphs	3	3	Additions
Addition of Points	9	9	
Addition of Clauses	6	12	
Repeal of Points	7	15	Repeals
Repeal of Clauses	2	2	
Repeal of Articles	1	8	
Total	128	340	

In Decree 123, the platform recognized 128 items requiring modification including various types of changes ranging from amendments, additions, to repeals of specific clauses and points. Based on those 128 items, the system updated 340 articles, clauses, and points. They have been modified, added, or repealed during the update process. Moreover, the update process successfully updated 518 triples out of a total of 1582 triples in the knowledge base of road traffic laws.

Example: In point b of clause 3, article 2, it is required to change as follows:

“b) Amend point c of clause 4, article 5 as follows:

“c) Operating a vehicle that does not meet the conditions for electronic toll collection (vehicle without a terminal tag

or with a terminal tag but the account balance is insufficient to pay when passing through the electronic toll lane) to enter the lane dedicated to electronic toll collection at toll stations;” replacing the old content: *“c) Operating a vehicle that does not meet the conditions for electronic toll collection to enter the lane dedicated to electronic toll collection at toll stations.”*

The new content clearly specifies the cases where the vehicle does not meet the toll collection conditions, making it easier for readers to understand and apply in practice. In the updating process, additional triples will be added as follows:

- **vehicle** → **without** → **terminal tag**
- **terminal tag** → **insufficient** → **payment**

This experiment illustrates the platform’s capability to effectively identify and update changes in legal documents, proving the feasibility and potential of the platform in supporting legal experts and regulatory agencies in promptly and accurately updating legal information.

V. CONCLUSIONS

This platform has proven effective in updating legal documents promptly and accurately with the latest legal documents. Analyzing and mapping changes, combined with updating the corresponding knowledge graph, ensures that all legal amendments are reflected accurately and timely. Moreover, this method avoids the need to update the entire knowledge base, focusing only on the relevant knowledge graph, thus saving time and resources. Experimental results demonstrate the platform’s ability to efficiently identify and process a large volume of changes.

In the future, the research focus on fully automating the knowledge update process within the platform under the guidance of domain experts. This automation will encompass the detection of newly released legal documents and the subsequent processing of these documents to ensure timely knowledge integration. Furthermore, the platform’s applicability will be expanded to encompass a wider range of domains, thereby enhancing its versatility.

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